

5G RAN

NG-flex Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

Technical Changes

None

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FBFD-021104	SA Option 2 Architecture	4 NG-flex

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
NG-flex	Supported only in SA networking	4 NG-flex

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
NG-flex	None	4 NG-flex

3 Overview

The standalone (SA) Option 2 network architecture includes the 5G Core Network (5GC) and the NG Radio Access Network (NG-RAN). The NG-RAN consists of gNodeBs, and the 5GC mainly consists of the access and mobility management function (AMF) and the user plane function (UPF). The gNodeB is connected to the AMF through the NG-C interface to implement NG control plane functions.

NG-flex allows NG-C connection establishment between a gNodeB and all AMFs in an AMF set on an SA network using the following methods:

- An NG-C connection is established between a gNodeB and an AMF, as shown in [Figure 3-1](#).
- Multiple NG-C connections are established between a gNodeB and an AMF. This connection mode is also called Transport Network Layer Association (TNLA) in 3GPP specifications, indicating the transmission association between a gNodeB and an AMF, as shown in [Figure 3-2](#).

When a UE accesses the network over the gNodeB, the gNodeB selects an AMF in the AMF set to serve the UE. If the gNodeB has established multiple TNLAs with the AMF, the gNodeB also needs to select a TNLA of the AMF to serve the UE.

After NG-flex is introduced, load balancing can be implemented in an AMF set to improve resource utilization.

Figure 3-1 AMF set

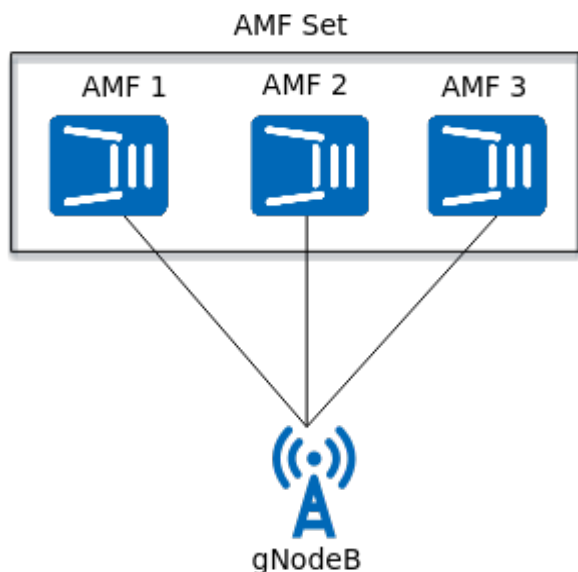
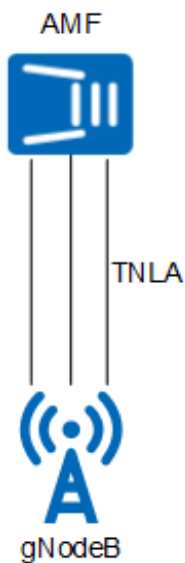


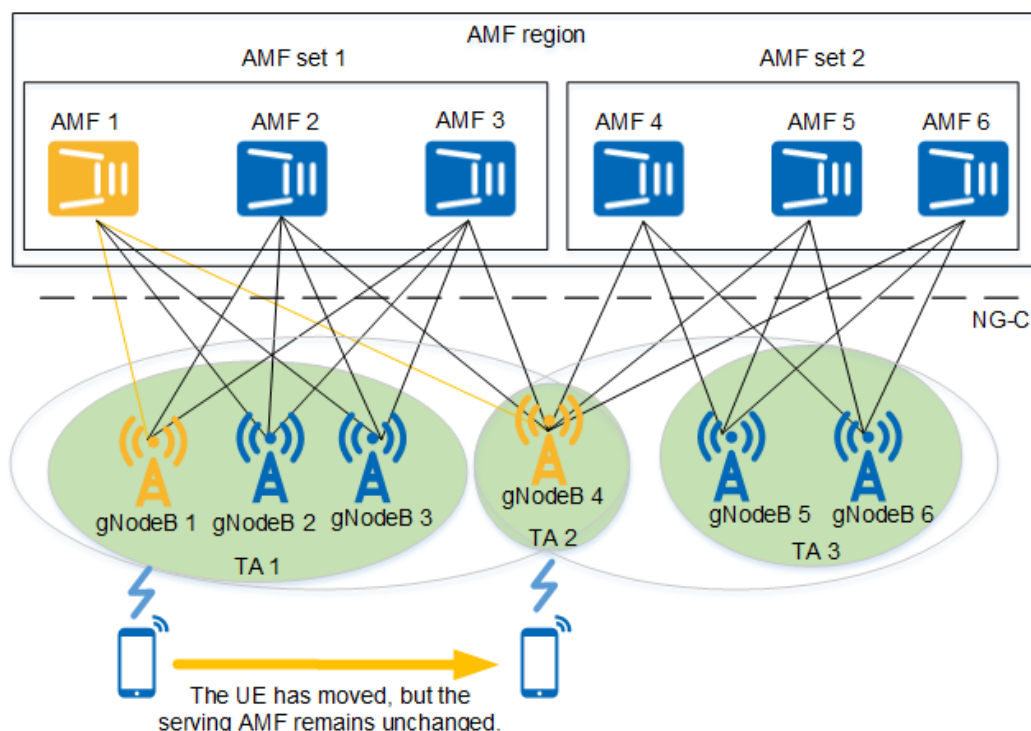
Figure 3-2 Multiple TNLAs between an AMF and a gNodeB



- Multiple transmission addresses can be planned on the AMF side. The gNodeB establishes a TNLA connection with each transmission address of the AMF.
- The AMF specifies the weight factor of each TNLA, and the gNodeB selects a TNLA based on the weight factor to implement load balancing.

NG-flex allows all AMFs in an AMF set to connect to all gNodeBs in a tracking area (TA). A TA is a geographic area composed of cells that provide continuous coverage. As shown in [Figure 3-3](#), AMF set 1 corresponds to TA 1 and TA 2, and AMF set 2 corresponds to TA 2 and TA 3.

Figure 3-3 AMF sets and TAs



- Each gNodeB in a TA connects to all AMFs in an AMF set.
- One AMF set corresponds to one or more TAs.
- When a UE moves among different TAs within an AMF set, the serving AMF remains the same.

The gNodeB distinguishes AMFs by their identifiers. The following describes related concepts.

AMF Set

An AMF set consists of several AMFs that serve a given area and network slice(s). All AMFs in an AMF set support the same network slice(s). For details about AMF sets, see section 3.1 "Definitions" and section 5.19 "Control Plane Load Control, Congestion and Overload Control" in 3GPP TS 23.501 V15.4.0.

AMF Region

An AMF region consists of one or more AMF sets.

GUAMI

In the 5GC, an AMF can be logically divided into multiple entities, which are identified by globally unique AMF identifiers (GUAMIs) in the following format: <MCC> <MNC> <AMF region ID> <AMF set ID> <AMF pointer>.

The *RegisteredAMF* IE carried in the RRCSetupComplete message sent from a UE to a gNodeB indicates the GUAMI, which is also known as the AMF identifier.

5G-S-TMSI

A 5G S-temporary mobile subscriber identity (5G-S-TMSI) is a UE's temporary identifier used to reduce the size of signaling messages over the air interface and improve the air interface efficiency. The 5G-S-TMSI is expressed in the following format: <AMF set ID> <AMF pointer> <5G-TMSI>.

AMF Name

An AMF name uniquely identifies an AMF. For details about the AMF name, see section 5.9.5 "AMF Name" in 3GPP TS 23.501 V15.4.0.

S-NSSAI

A Single Network Slice Selection Assistance Information (S-NSSAI) contains such information as the slice/service type. For details, see section 5.15.2 "Identification and selection of a Network Slice: the S-NSSAI and the NSSAI" in 3GPP TS 23.501 V15.4.0.

NOTE

A UE-provided GUAMI, 5G-S-TMSI, or S-NSSAI can be used as the identifier for selecting an AMF.

4 NG-flex

4.1 Principles

For a gNodeB, NG-flex selects an AMF for a UE based on the following methods:

- Selecting an AMF
 - If the UE provides an AMF ID, the gNodeB preferentially selects the AMF for the UE based on this ID.
 - If the UE does not provide an AMF ID or no appropriate AMF is selected based on the AMF ID provided by the UE:
 - If priority-based AMF selection is disabled, the gNodeB selects an AMF based on the relative capacity. For details, see [AMF Selection Based on the Relative Capacity](#).
 - If priority-based AMF selection is enabled, the gNodeB selects an AMF based on the AMF priority and then based on the relative capacity. For details, see [AMF Selection Based on the AMF Priority](#).
- Selecting a TNLA

The gNodeB selects a TNLA based on the weight factor. For details, see [4.1.2 Selecting a TNLA](#).
- AMF load balancing

The load is balanced among all AMFs in an AMF set. For details, see [4.1.3 AMF Load Balancing](#).
- AMF overload processing

Upon receipt of an Overload Start message from an AMF, the gNodeB processes the message based on the policy specified in the message. For details, see [4.1.4 AMF Overload Processing](#).
- AMF troubleshooting

When an AMF is faulty, the gNodeB can detect the faulty AMF and process calls under the faulty AMF. For details, see [4.1.5 AMF Troubleshooting](#).

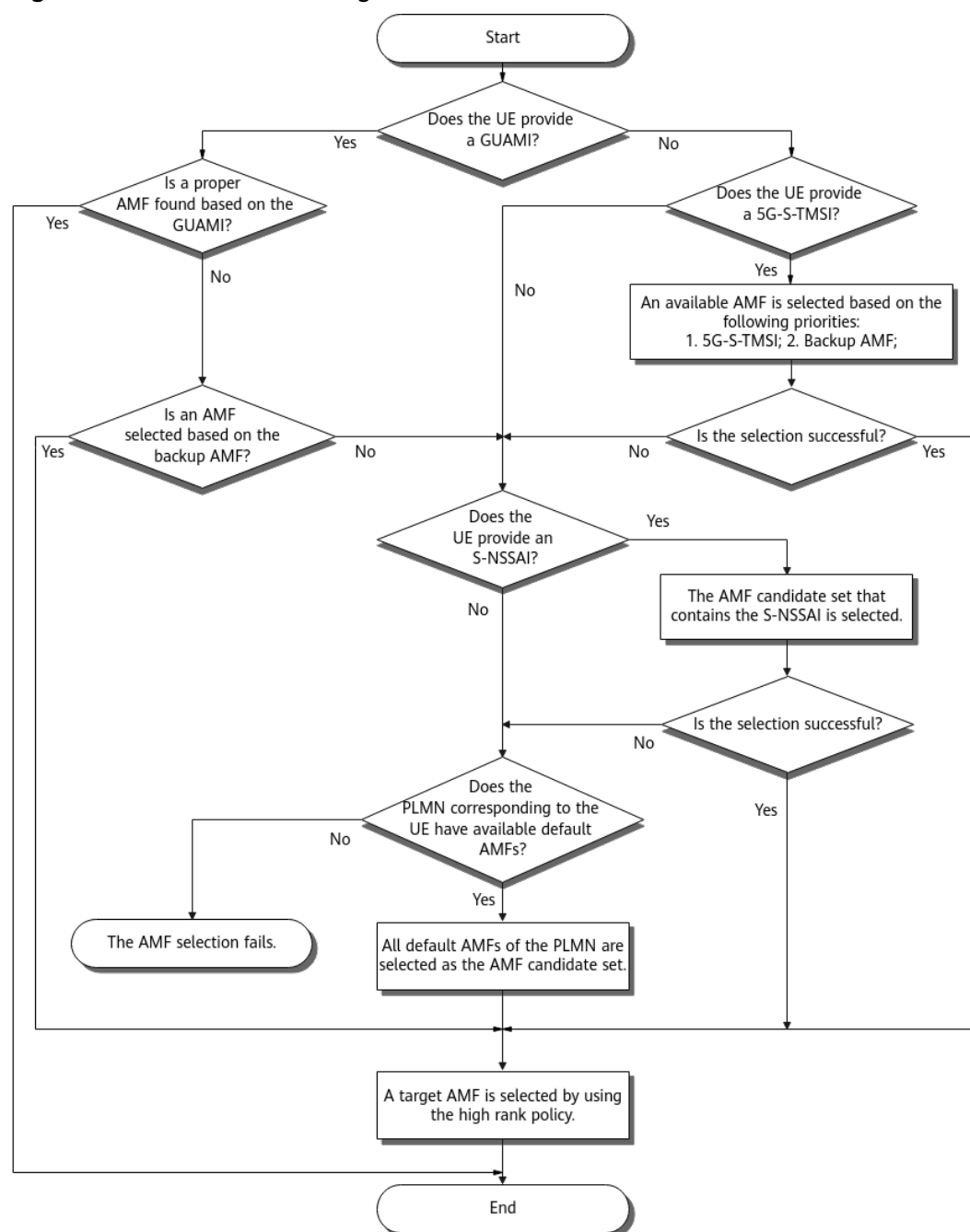
4.1.1 Selecting an AMF

4.1.1.1 Selecting an AMF

When a UE accesses a gNodeB, the gNodeB needs to establish a dedicated connection to an AMF. If the gNodeB is connected to multiple AMFs, the gNodeB selects an AMF for the UE to establish a dedicated connection. When the gNodeB receives an RRCSetupComplete message that contains an AMF ID from the UE, the gNodeB matches the AMF ID against the AMF information stored on the gNodeB and selects the corresponding AMF.

Figure 4-1 shows the process of selecting an AMF.

Figure 4-1 Process of selecting an AMF



- Whether the UE provides a GUAMI
 - If a GUAMI is provided, further determination is performed based on whether a proper AMF can be found based on the GUAMI.
 - If a desired AMF is found, the gNodeB selects the AMF and sets up a connection with it.
 - If no desired AMF is available (for example, the found AMF is faulty), an AMF is preferentially selected from backup AMFs based on the information contained in the GUAMI.
 - If the selection is successful, a target AMF is selected by using the high rank policy and a connection is set up with the target AMF. For details, see [4.1.1.2 AMF Selection Based on the High Rank Policy](#).
 - If the selection fails, further determination is required based on whether the UE provides an S-NSSAI.
 - If a GUAMI is not provided, further determination is required based on whether the UE provides a 5G-S-TMSI.
- Whether the UE provides a 5G-S-TMSI
 - If a 5G-S-TMSI is provided, the gNodeB preferentially selects an AMF from the 5G-S-TMSI and then from backup AMFs based on the information contained in the 5G-S-TMSI.
 - If the selection is successful, a target AMF is selected by using the high rank policy and a connection is set up with the target AMF. For details, see [4.1.1.2 AMF Selection Based on the High Rank Policy](#).
 - If the selection fails, further determination is required based on whether the UE provides an S-NSSAI.
 - If a 5G-S-TMSI is not provided, further determination is required based on whether the UE provides an S-NSSAI.
- Whether the UE provides an S-NSSAI
 - If an S-NSSAI is provided, the gNodeB selects the AMF candidate set that contains the S-NSSAI.
 - If the selection is successful, a target AMF is selected by using the high rank policy and a connection is set up with the target AMF. For details, see [4.1.1.2 AMF Selection Based on the High Rank Policy](#).
 - If the selection fails, further determination is required based on whether the PLMN corresponding to the UE has available default AMFs.
 - If an S-NSSAI is not provided, further determination is required based on whether the PLMN corresponding to the UE has available default AMFs.
- Whether the PLMN corresponding to the UE has available default AMFs
 - If yes, the gNodeB selects all default AMFs of the PLMN as the candidate AMF set, uses the high rank policy to select the target AMF, and sets up a connection with the target AMF. For details, see [4.1.1.2 AMF Selection Based on the High Rank Policy](#).
 - If no, the AMF selection fails.

The details are as follows:

- The **gNBCUNg** MO is configured to specify default AMFs. The **gNBCUNg.gNBCuNgId** parameter is configured to specify the NG interface to be associated with a default AMF, and the **gNBCUNg.CuNgType** parameter must be set to **DEFAULT_CP** or **DEFAULT_CP_UP**. As a default AMF is operator specific, it needs to be configured for each operator in multi-operator scenarios. The operator information of a default AMF is configured using the **gNBCUNgOp** MO. If no default AMFs are configured for an operator, all available normal AMFs of the operator are considered as available default AMFs.
- An AMF candidate set may include a plurality of AMF sets.
- You can run the **DSP GNBCUNGINTERFACE** command to query the served GUAMI of the AMF and the corresponding backup AMF information.

NOTE

Assume that a UE in connected mode has established a connection with an AMF. If the AMF is faulty, the gNodeB allows the UE to establish a connection with the backup AMF corresponding to the faulty AMF. Then, the uplink NG-C messages of the UE are sent to the backup AMF, and the downlink NG-C messages are also received by the backup AMF. For details, see section 5.21.2.3 "Procedure for AMF Auto-recovery" in 3GPP TS 23.501 V16.8.0 and section 4.2.7.2.4 "NGAP UE-TNLA-binding update procedure" in 3GPP TS 23.502 V16.8.0.

4.1.1.2 AMF Selection Based on the High Rank Policy

If a UE does not provide an AMF ID (that is, the RRCSetupComplete message does not contain an AMF ID) or the AMF ID provided by the UE cannot be used to select a proper AMF, the gNodeB needs to select a proper AMF from the AMF candidate set based on the high rank policy to provide services for the UE. The process of selecting an AMF varies depending on whether priority-based AMF selection (specified by the **AMF_PRIORITY_SELECT_SW** option of the **gNodeBParam.NgInterfaceAlgoSw** parameter) is enabled. If priority-based AMF selection is disabled, the gNodeB selects an AMF based on the relative capacity of the core network, as detailed in [AMF Selection Based on the Relative Capacity](#). If priority-based AMF selection is enabled, the gNodeB selects an AMF based on the AMF priority, as detailed in [AMF Selection Based on the AMF Priority](#), and then based on the relative capacity.

AMF Selection Based on the Relative Capacity

If priority-based AMF selection is disabled (by deselecting the **AMF_PRIORITY_SELECT_SW** option of the **gNodeBParam.NgInterfaceAlgoSw** parameter), the gNodeB selects an AMF from the candidate AMF set based on the relative capacity of the core network. For example, if the relative capacity ratio of AMF 1 to AMF 2 is 2:1, the possibility of AMF 1 being selected is twice that of AMF 2 based on the algorithm. Based on the probability statistics theories, when more UEs access the network, the result of the algorithm will be closer to the relative AMF capacity ratio. The **DSP GNBCUNGINTERFACE** command can be executed to query the relative AMF capacity.

The gNodeB establishes NG-AMF connections with all AMFs in an AMF candidate set and maintains the relative capacity of each AMF as follows:

- When the gNodeB is establishing an NG-AMF connection with an AMF:
The AMF sends the gNodeB an NG SETUP RESPONSE message that carries the **Relative AMF Capacity** field, notifying the gNodeB of the relative AMF capacity.
- After the gNodeB establishes an NG-AMF connection with the AMF:
If the processing capability of the AMF is updated, the AMF sends the gNodeB an AMF CONFIGURATION UPDATE message that carries the **Relative AMF Capacity** field, notifying the gNodeB of the updated relative AMF capacity.

 NOTE

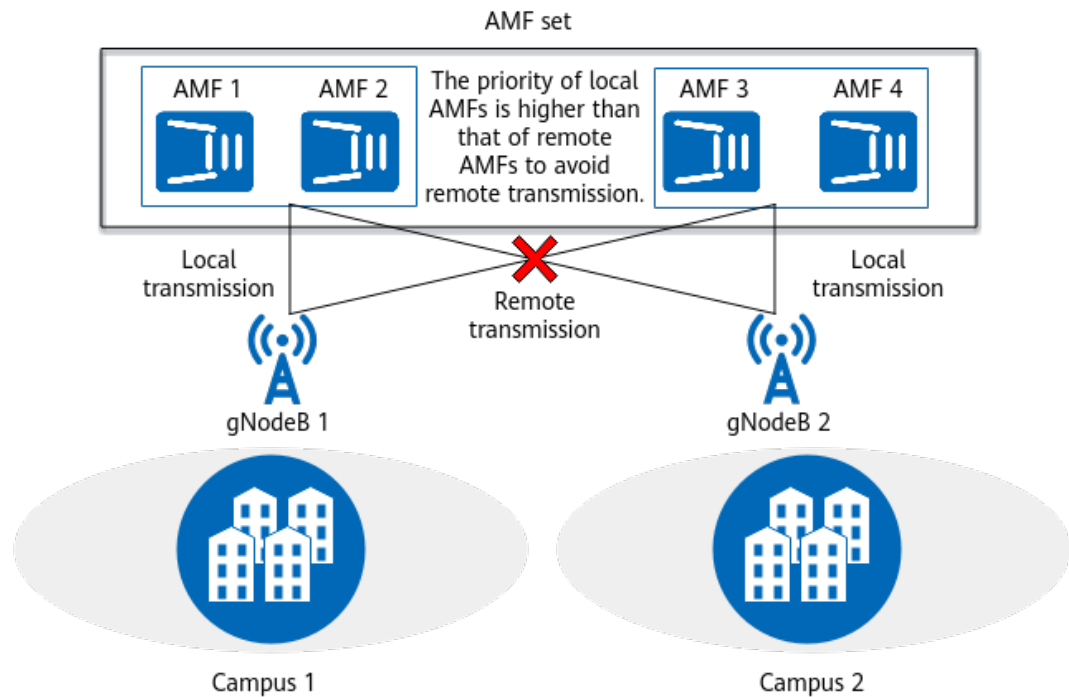
In some cases, the relative AMF capacity may not be set based on the actual processing capability. For example, if an AMF is added to an AMF candidate set or the capacity of an original AMF is expanded, the relative AMF capacity can be set to a value that exceeds the actual processing capability to rapidly increase the AMF load. After the AMF load increases to a desired level, the relative capacity can be changed to the actual value.

The gNodeB selects an AMF based on the **Relative AMF Capacity** value. The gNodeB may not evenly allocate UEs across AMFs even if AMFs have the same **Relative AMF Capacity** value.

AMF Selection Based on the AMF Priority

AMFs are backed up in multiple locations for redundancy. When the gNodeB selects an AMF based on the relative capacity of the core network, the gNodeB may access a remote AMF, as shown in [Figure 4-2](#). If the gNodeB accesses a remote AMF, data will be transmitted remotely. (In ToB scenarios, data will be transmitted out of campus.) In addition, unstable remote transmission decreases the AMF access success rate. If priority-based AMF selection is enabled (by selecting the **AMF_PRIORITY_SELECT_SW** option of the **gNodeBParam.NgInterfaceAlgoSw** parameter), the gNodeB selects an AMF from the candidate AMF set based on the AMF priority and then based on the relative capacity. For example, if the priority of AMF 1 and AMF 2 is set to 100 and the priority of AMF 3 and AMF 4 is set to 50, gNodeB 1 selects AMF 1 and AMF 2 as the candidate set based on the priority-based AMF selection algorithm, as a larger value indicates a higher priority (absolute priority). Then further selection is performed based on the relative capacity (detailed in [AMF Selection Based on the Relative Capacity](#)). When local AMFs are configured with a higher priority and remote AMFs are configured with a lower priority on the gNodeB, enabling priority-based AMF selection ensures that the gNodeB preferentially accesses local AMFs with a higher priority. The gNodeB accesses remote AMFs with a lower priority only when all local AMFs are unavailable.

Figure 4-2 AMF priorities



The AMF priority list is configured using the parameters in the **GNODEBAMFCONFIG** MO.

- The **gNodeBAmfConfig.AmfConfigIndex** parameter specifies the AMF configuration index. An AMF parameter combination is uniquely identified by this parameter.
- The **gNodeBAmfConfig.Mcc** parameter specifies the mobile country code (MCC) of a PLMN. The MCC consists of three digits.
- The **gNodeBAmfConfig.Mnc** parameter specifies the mobile network code (MNC) of a PLMN. The MNC consists of two or three digits.
- The **gNodeBAmfConfig.AmfName** parameter specifies the AMF name.
- The **gNodeBAmfConfig.AmfPriority** parameter specifies the AMF priority for the gNodeB. A larger value indicates a higher priority.

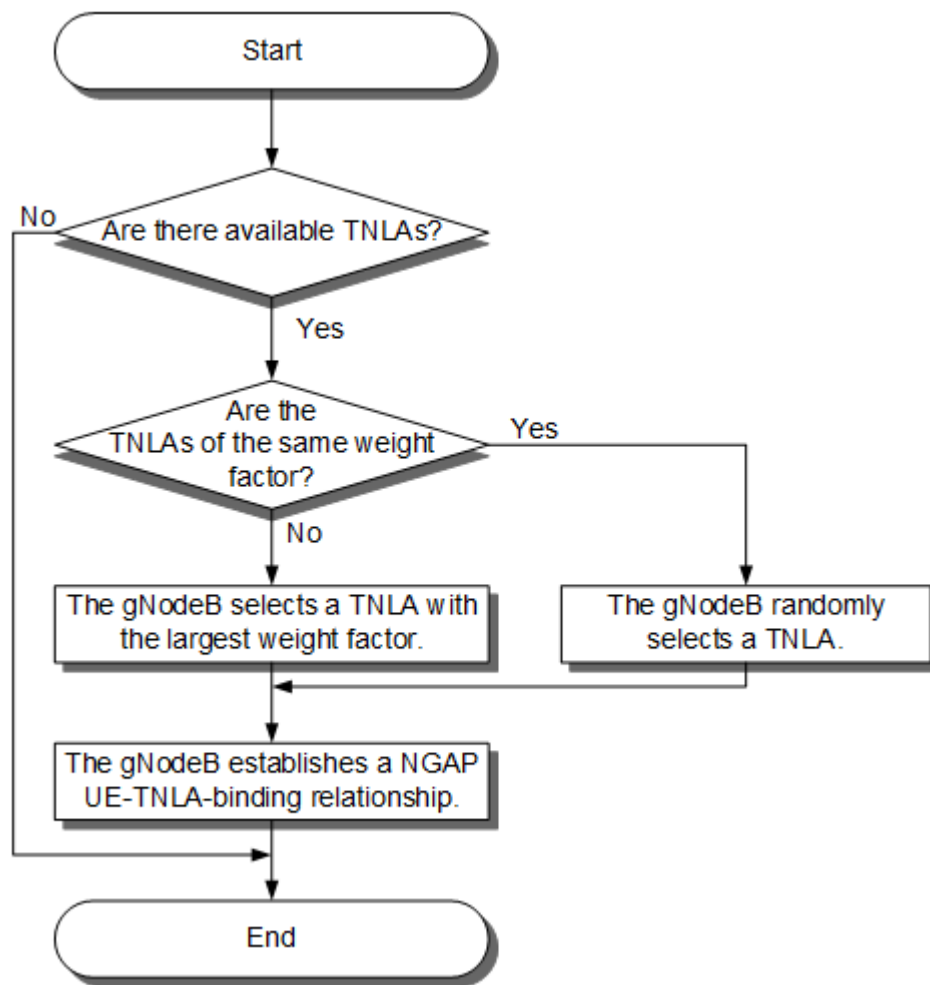
4.1.2 Selecting a TNLA

According to section 5.21.1.3 "N2 TNL association selection" in 3GPP TS 23.501 V15.4.0, after a gNodeB selects an AMF with multiple TNLAs for a UE, the gNodeB selects a TNLA mainly based on:

- TNLA availability (as indicated by the **usage** field)
- TNLA weight factor (as indicated by the **weight factor** field)

Figure 4-3 shows the process of selecting a TNLA.

Figure 4-3 Process of selecting a TNLA



- The gNodeB selects a TNLA of an AMF based on the weight factor.
For example, if weight factors of TNLA 1 and TNLA 2 are 2 and 1, respectively, the possibility of TNLA 1 being selected is twice that of TNLA 2 based on the algorithm. Based on the probability statistics theories, when more UEs access the network, the final algorithm effect will be closer to the weight factor ratio.

NOTE

The gNodeB selects a TNLA based on the weight factor, but does not ensure that UEs are evenly allocated across TNLAs.

- The AMF can set the TNLA availability to UE only, non-UE only, or both.
When a UE accesses the network, only the TNLA links whose availability is set to UE-only or both can be selected. For details, see section 8.7.3 "AMF Configuration Update" in 3GPP TS 38.413 V15.2.0.
- The gNodeB establishes a TNLA connection with the AMF and obtains the TNLA availability and weight factor as follows:
 - When the gNodeB is establishing a TNLA connection with an AMF:
The AMF sends the gNodeB an AMF CONFIGURATION UPDATE message that carries the **usage** (optional) and **weight factor** fields in the *AMF TNL Association to Add Item* IE.

- After the gNodeB establishes a TNLA connection with an AMF:
If the processing capability of the TNLA is updated, the AMF sends the gNodeB the AMF CONFIGURATION UPDATE message that carries the **usage** (optional) and **weight factor** fields in the *AMF TNL Association to Update Item* IE.

4.1.3 AMF Load Balancing

AMF load balancing adjusts the load across AMFs by transferring the UE context information registered on an AMF to other AMFs. An AMF can request some or all of the serving gNodeBs to redirect a UE that has changed from the RRC_IDLE state to the RRC_CONNECTED state to another AMF in the AMF set.

For UEs in either the RRC_IDLE or RRC_CONNECTED state, AMF load balancing works as follows:

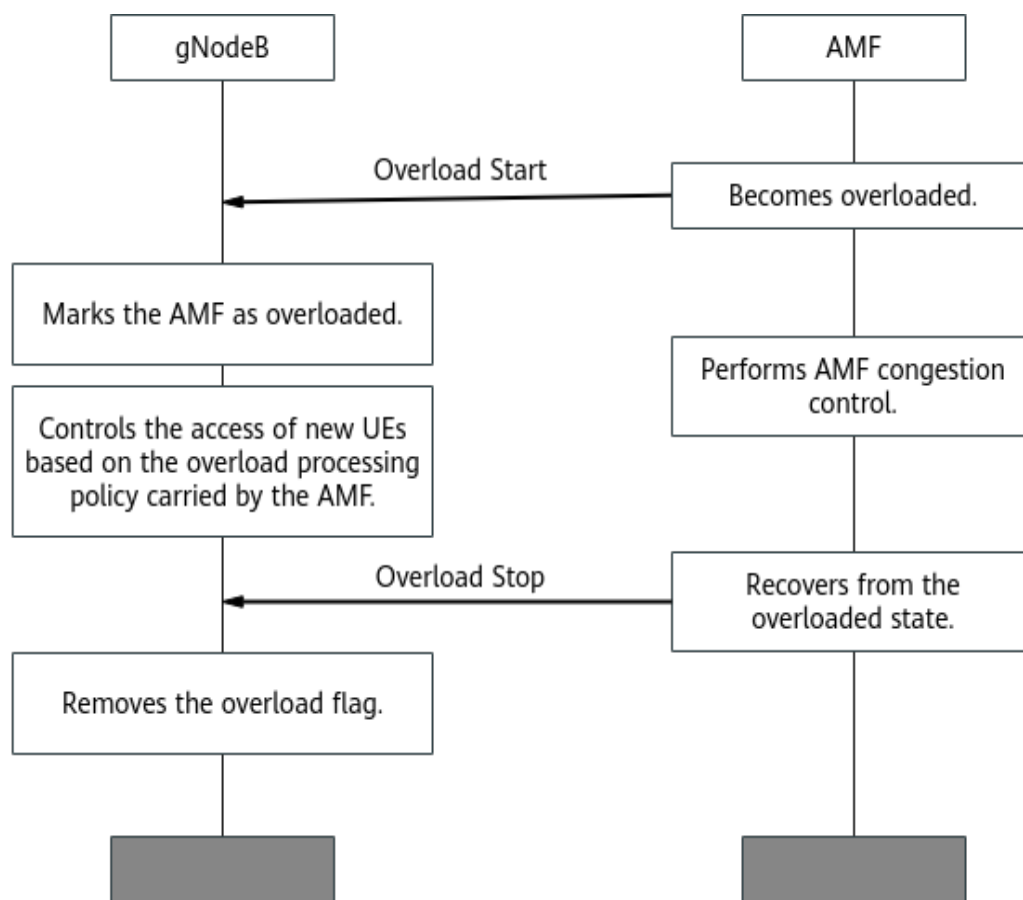
- When a UE shifts from the RRC_IDLE state to the RRC_CONNECTED state and the gNodeB receives an initial NAS message requesting redirection from the AMF, the gNodeB selects the target AMF based on the information (5G S-TMSI or GUAMI) carried in the message.
- For a UE in the RRC_CONNECTED state, AMF load balancing can be implemented by the AMF management function. For details, see section 5.21.2 "AMF Management" in 3GPP TS 23.501 V15.4.0.

The AMF Status Indication field notifies the gNodeB of unavailable AMFs and their backup AMFs (which are optional). When the gNodeB needs to send an uplink message, it preferentially selects a backup AMF. If no backup AMF is available, the gNodeB reselects an AMF according to [4.1.1 Selecting an AMF](#).

4.1.4 AMF Overload Processing

[Figure 4-4](#) shows the interaction between an AMF and a gNodeB when the AMF is overloaded.

Figure 4-4 Interaction between a gNodeB and an AMF for overload control



1. When the AMF becomes overloaded, it sends an Overload Start message to the gNodeB.
2. Upon receipt of this message, the gNodeB marks the AMF as overloaded.
3. Both the gNodeB and AMF attempt to alleviate the AMF load.
4. After the AMF is no longer overloaded, it sends an Overload Stop message to the gNodeB.
5. Upon receipt of this message, the gNodeB removes the overload flag from the AMF, updates the corresponding AMF set information, and resumes processing as normal.

Reducing the AMF Load on the gNodeB

An AMF sends an Overload Start message to instruct the gNodeB to either allow or reject UE calls of a specific type and to specify the proportion of UEs whose calls are allowed or rejected. The gNodeB determines whether to allow or reject UE access to the overloaded AMF based on the instruction.

NOTE

Upon receipt of an Overload Start message from an AMF, the gNodeB stops allocating new UEs matching the overload cause to the AMF. However, UEs already served by the AMF may still increase the load of the AMF.

Table 4-1 exemplifies the mappings between the gNodeB processing policies carried in the Overload Start message from an AMF and the reasons for UE access carried in the RRCSetupRequest message.

Table 4-1 Examples of mapping between the gNodeB processing policies and UE access reasons

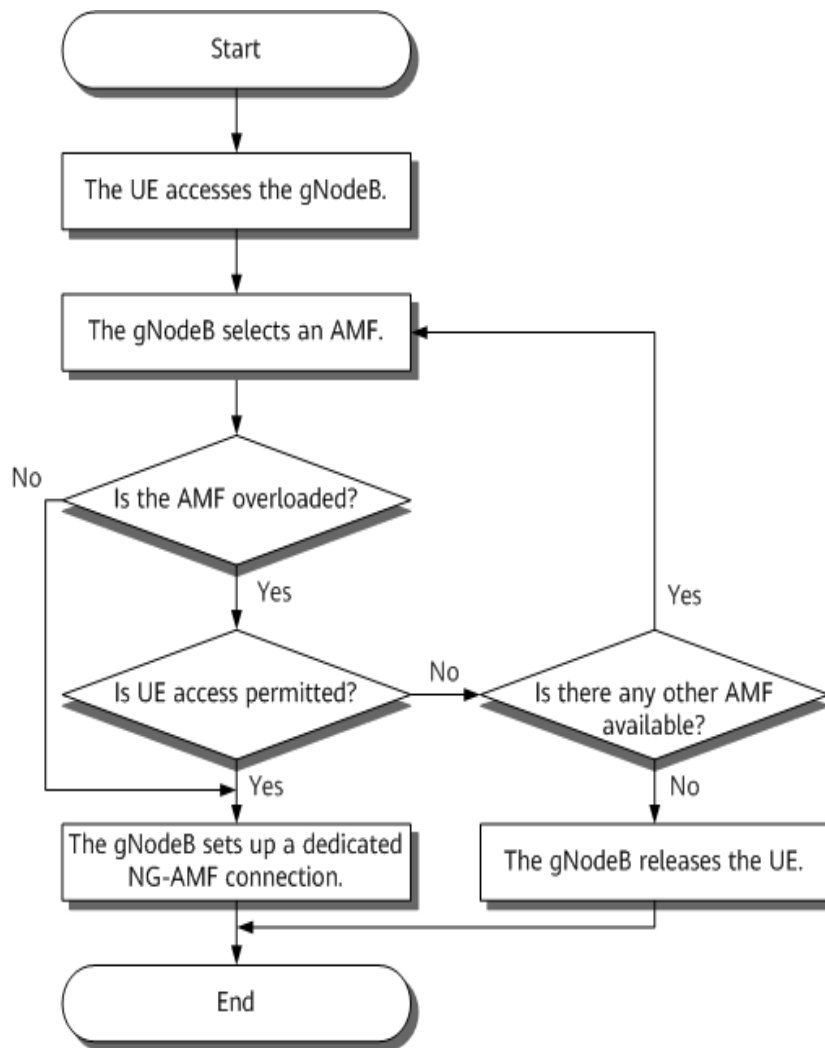
Overload Processing Policy Carried in the AMF Overload Start Message	Reason for UE Access
Reject RRC connection establishments for non-emergency MO DT	If the UE initiates an RRC connection setup request with the cause value of mo-Data , mo-SMS , mo-VideoCall , or mo-VoiceCall , the gNodeB rejects the request.
Reject RRC connection establishments for Signalling	If the UE initiates an RRC connection setup request with the cause value of mo-Data , mo-SMS , mo-Signalling , mo-VideoCall , or mo-VoiceCall , the gNodeB rejects the request.
Permit Emergency Sessions and mobile terminated services only	If the UE initiates an RRC connection setup request with the cause value of emergency or mt-Access , the UE access is permitted.
Permit High Priority Sessions and mobile terminated services only	If the UE initiates an RRC connection setup request with the cause value of highPriorityAccess or mt-Access , the UE access is permitted.

 NOTE

For details about the overload processing policies carried in the Overload Start message from an AMF, see section 8.7.7 "Overload Start" in 3GPP TS 38.413 V15.2.0. For details about the reasons for UE access carried in the RRCSetupRequest message, see section 5.3.3 "RRC connection establishment" in 3GPP TS 38.331 V15.4.0.

Figure 4-5 shows the AMF selection procedure when an AMF is overloaded

Figure 4-5 AMF selection procedure when an AMF is overloaded



1. The UE accesses the gNodeB and requests to set up an RRC connection.
2. The gNodeB selects an AMF based on the UE-provided AMF identifier and AMF capacity.
3. The gNodeB checks whether the AMF is overloaded.
 - If the AMF is overloaded:
 - If the requested service should be allowed, the gNodeB selects the AMF and proceeds to 5.
 - If the requested service should be rejected, the gNodeB proceeds to 4.
 - If the AMF is not overloaded, the gNodeB proceeds to 5.
4. The gNodeB checks whether there are other available AMFs.
 - If other AMFs exist, the gNodeB selects one and returns to 2.
 - If no other AMF exists, the gNodeB releases the UE.
5. The gNodeB sets up a dedicated NG-AMF connection.

4.1.5 AMF Troubleshooting

The following two types of fault may be the cause:

- If the gNodeB detects that all TNLAs of the AMF are faulty, the AMF is faulty. (The AMF status can be queried by running the **DSP GNBCUNGINTERFACE** command.)
- The gNodeB receives a status information from the core network, indicating that there are no available GUAMIs.

When the AMF is faulty, the gNodeB performs the following operations:

- For a UE that newly accesses the network:
 - If its home AMF is faulty and a backup AMF is available for the home AMF, the backup AMF is selected to establish an NG-C connection.
 - If no backup AMF is available for the home AMF or the backup AMF is also faulty, an AMF working normally in the same set as the home AMF is selected to establish an NG-C connection.
- For an online UE:
 - If a backup AMF is available for the faulty AMF, the gNodeB selects the backup AMF to restore the NG-C connection when the gNodeB needs to send uplink messages or receives downlink messages from the backup AMF.
 - If no backup AMF is available for the faulty AMF or the backup AMF is also faulty, check whether the following conditions are met:
 - The **AMF_SET_BACKUP_SW** option of the **gNodeBParam.NgInterfaceAlgoSw** parameter is selected.
 - An AMF in the same set as the faulty AMF is working properly.

If all the preceding conditions are met, the gNodeB selects the AMF that is working properly in the same set as the faulty AMF to restore the NG-C connection. Otherwise, the gNodeB releases all UEs served by the faulty AMF.

4.2 Network Analysis

4.2.1 Benefits

NG-flex provides the following benefits:

- When a UE moves within the coverage area of the same AMF set, the serving AMF does not need to change. This reduces the signaling overhead arising from location area update and improves the system processing capability.
- Load balancing in the AMF set improves resource utilization.
- Adding and deleting AMF nodes is an easier method to minimize the impact on ongoing services.
- Inter-AMF mutual backup improves network reliability.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

When priority-based AMF selection is enabled and local AMFs are prioritized over remote AMFs, the gNodeB can preferentially access local AMFs with higher priorities. This prevents the gNodeB from accessing remote AMFs with lower priorities and increases the access success rate.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards support this function. To learn which main control boards are NR-capable, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Others

The core network must support NG-flex. NG-flex is not a Huawei proprietary function and can be used with core network devices of any vendor. Non-Huawei core network devices that support this function can also be used. An AMF needs to support the configuration of the AMF pool (one AMF set corresponds to one AMF pool) and send the IEs required by NG-flex, such as *AMF Region*, *AMF Set*, and *Relative AMF Capacity*, to the gNodeB during the setup of the NG-C connection. NG-flex has restrictions on AMF and TA planning. For details, see [4.1 Principles](#).

Assuming that networking requirements are met, a list of AMFs to be connected to each gNodeB is required, and the following must be configured: corresponding SCTP links, NG interfaces, transport layer objects, and corresponding route information.

According to networking requirements, an operator needs to plan one or more IP addresses and SCTP port numbers for the NG-C interface and commission the transmission link between the gNodeB and the AMF.

4.4 Operation and Maintenance

4.4.1 When to Use

- It is recommended that this function be enabled when the core network supports NG-flex and an AMF pool is configured.
- When the **gNBCUNg** MO is used to configure the default AMF, the following requirements must be met:
In live network planning, if a **gNBCUNg** MO of the default type (**gNBCUNg.CuNgType** set to **DEFAULT_CP** or **DEFAULT_CP_UP**) is configured, there must be an available NG link under the **gNBCUNg** MO of the default type. Otherwise, services will be affected.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

NG-flex requires multiple SCTP peers to be configured for the NG interface. For details about data preparation for NG interface configuration, see *NG and Xn Self-Management*. For other required data, see [Table 4-2](#).

Table 4-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Ng Interface Algorithm Switch	gNodeBParam.NgInterfaceAlgoSw	AMF_SET_BACKUP_SW	To support AMF backup in an AMF set for UEs in connected mode in SA networking, select the corresponding option.
AMF Configuration Index	gNodeBAmfConfig.AmfConfigIndex	None	Set this parameter based on the network plan.
Mobile Country Code	gNodeBAmfConfig.Mcc	None	Set this parameter based on the network plan.
Mobile Network Code	gNodeBAmfConfig.Mnc	None	Set this parameter based on the network plan.
AMF Name	gNodeBAmfConfig.AmfName	None	Set this parameter based on the network plan.
AMF Priority	gNodeBAmfConfig.AmfPriority	None	Set this parameter based on the network plan. A larger value indicates a higher priority.
Ng Interface Algorithm Switch	gNodeBParam.NgInterfaceAlgoSw	AMF_PRIORITY_SELECT_SW	If the gNodeB needs to select an AMF based on the AMF priority, configure the AMF priority list (by configuring parameters in the GNODEBAMFCONFIG MO) and then select this option.

Parameter Name	Parameter ID	Option	Setting Notes
CU NG Type	gNBCUNg.CuNgType	None	Set this parameter based on the network plan. Set this parameter to DEFAULT_CP or DEFAULT_CP_UP when configuring a default AMF. Set this parameter to DEFAULT_UP or DEFAULT_CP_UP when configuring a default UPF. When the control plane and user plane are configured on two NG interfaces respectively, set this parameter to DEFAULT_CP for one NG interface and DEFAULT_UP for the other. When the control plane and user plane are configured on one NG interface, set this parameter to DEFAULT_CP_UP.

4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- Step 1** Run the **ADD SCTPTEMPLATE** command to add an SCTP parameter template. Multiple SCTP links can share one SCTP parameter template.
- Step 2** Run the **ADD SCTPHOST** command to add an SCTP host. Multiple SCTP peers can correspond to one SCTP host.

- Step 3** Run the **ADD SCTPPEER** command to add an SCTP peer. An SCTP peer needs to be added for each AMF.
- Step 4** Run the **ADD EPGROUP** command to add an endpoint group. Different AMFs can be configured with the same endpoint group. However, if AMFs are not shared among operators, different endpoint groups must be configured. For details, see *Multi-Operator Sharing*.
- Step 5** Run the **ADD SCTPHOST2EPGRP** command to add the SCTP host to the endpoint group corresponding to the AMF to be connected.
- Step 6** Run the **ADD SCTPPEER2EPGRP** command to add the SCTP peer to the endpoint group.
- Step 7** Run the **ADD GNBCUNG** command to add an NG object and associate it with the control-plane endpoint group.

----End

Activation Command Examples

Control-plane configuration

```
//Adding an SCTP parameter template
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an SCTP host with the first local IP address (SCTPHOST.SIGIP1V4) the same as the IP address of
the gNodeB (IPv4 used as an example)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="70.190.103.8", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=2911, SCTPTEMPLATEID=0, USERLABEL="for AMF";
//Adding SCTP peers with the first peer IP address (SCTPPEER.SIGIP1V4) the same as the IP address of the
core network (IPv4 used as an example)
ADD SCTPPEER: SCTPPEERID=0, IPVERSION=IPv4, SIGIP1V4="30.0.0.20", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=1032, USERLABEL="for AMF1";
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="30.0.0.38", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=1030, USERLABEL="for AMF2";
ADD EPGROUP: EPGROUPID=0, USERLABEL="for NG";
ADD SCTPHOST2EPGRP: EPGROUPID=0, SCTPHOSTID=0;
ADD SCTPPEER2EPGRP: EPGROUPID=0, SCTPPEERID=0;
ADD SCTPPEER2EPGRP: EPGROUPID=0, SCTPPEERID=1;
```

User-plane configuration (NG-flex involves only the control-plane configuration. The user-plane configuration is a part of the basic function configuration of the NG interface.)

```
//Adding a user-plane host (IPv4 used as an example)
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="70.190.103.1", IPSECSWITCH=DISABLE,
USERLABEL="for UPF";
//Adding the user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
//Adding a gNodeB CU NG object in either of the following scenarios
//With a default AMF not configured
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0;
//With a default AMF configured
ADD GNBCUNG: gNBCuNgId=0, CpEpGroupId=0, UpEpGroupId=0, CuNgType=DEFAULT_CP;
//Selecting the AMF_SET_BACKUP_SW option
MOD GNODEBPARAM: NgInterfaceAlgoSw=AMF_SET_BACKUP_SW-1;
//Adding parameters related to the AMF priority configuration
ADD GNODEBAMFCONFIG: AmfConfigIndex=0, Mcc="460", Mnc="06", AmfName="1", AmfPriority=0;
//Modifying parameters related to the AMF priority configuration
MOD GNODEBAMFCONFIG: AmfConfigIndex=0, Mcc="450", Mnc="05", AmfName="2", AmfPriority=255;
//Enabling priority-based AMF selection
MOD GNODEBPARAM: NgInterfaceAlgoSw=AMF_PRIORITY_SELECT_SW-1;
```

4.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.3 Activation Verification

NG Interface Status

Step 1 Run the **DSP GNBCUNGINTERFACE** command to query the status of all NG interfaces configured on the gNodeB.

- If **gNodeB CU NG Interface State** is **Normal**, the NG interface is in the normal state.
- If **gNodeB CU NG Interface CP Bearer State** is **Normal**, the SCTP link is in the normal state.

If the status of all NG interfaces and SCTP links is normal, NG-flex has been activated.

Step 2 Run the **DSP GNBCUNGINTERFACE** command to query the value of **AMF UE Num** for each NG interface. If the UE access complies with the principles described in [4.1.1.1 Selecting an AMF](#), UEs served by AMFs are evenly distributed on each AMF. This indicates that the gNodeB performs load balancing across AMFs when selecting an AMF for UE access, and that NG-flex is in normal state.

----End

Relative AMF Capacity

Step 1 Run the **DSP GNBCUNGINTERFACE** command to query the relative AMF capacity.

Step 2 Enable a UE to access the gNodeB and check whether the UE access complies with the principle of selecting an AMF based on the relative capacity. For details, see [4.1.1.1 Selecting an AMF](#).

If it does, the gNodeB selects an AMF for the UE.

----End

4.4.4 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 23.501: "System Architecture for the 5G System"
- 3GPP TS 23.502: "Procedures for the 5G System"
- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *NG and Xn Self-Management*
- *Multi-Operator Sharing*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*